

Chapter 3. Lighting

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a X.Y notation where X is the chapter and Y is the sub-section within that chapter. References to external documents are contained in Chapter 1 and are made using [Rn] notation.

3.1. General Description

3.1.1. Introduction

The clusters specified in this document are for use typically in lighting applications, but MAY be used in any application domain.

3.1.2. Terms

HSV: Hue, Saturation, Value. A color space, also known as HSB (Hue, Saturation, Brightness). This is a well-known transformation of the RGB (Red, Green, Blue) color space. For more information see e.g., http://en.wikipedia.org/wiki/HSV_color_space.

Illuminance: The density of incident luminous flux on a surface. Illuminance is the standard metric for lighting levels, and is measured in lux (lx).

3.1.3. The Dimming Light Curve

The dimming curve is recommended to be logarithmic, as defined by the following equation:

$$\%Light = 10^{\left(\frac{Level-1}{\frac{253}{3}} \right) - 1}$$

Where: %Light is the percent light output of the ballast and Level is an 8-bit integer between 1 (0.1% light output) and 254 (100% output) that is adjusted for MinLevel and MaxLevel using the following equation:

$$Level = MinLevel + (MaxLevel - MinLevel) * (CurrentLevel - 1) / 253.$$

NOTE Value 255 is not used.

The table below gives a couple of examples of the dimming light curve for different values of MinLevel, MaxLevel and CurrentLevel.

Table 6. Examples of The Dimming Light Curve

MinLevel	MaxLevel	CurrentLevel	Level	%Light
1	254	1	1	0.10%
1	254	10	10	0.13%